

Supplemental Material for “Non-Additive Strain–Doping Coupling in Quasi-Hexagonal C₆₀ Graphullerene”

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OVERVIEW

This Supplemental Material accompanies the main PRB article on coupled (ϵ, δ) response in qHP C₆₀. All figures and tables are labeled with prefix **S** to match citations in the main text (Fig. S4–S6; Tables S1–S5). Calculations use the same PBE+DFT-D3 CP2K protocol as the main manuscript (rigid biaxial strain unless noted). Observables follow Capobianco *et al.* [1]: π -projected DOS, inverse participation ratio (IPR), nearest-neighbor coupling J , and Marcus reorganization energies $\lambda^\pm$. Marcus $\lambda^\pm$ in Fig. S5 follow the adiabatic plus vertical protocol of Ref. 1 at the main-text PBE+D3 level (12/12 charged geometry optimizations converged at revision; eight vertical single-points; 6/8 converged at revision). Converged inputs, outputs, and analysis JSON are archived as described in the main-text Data Availability statement (charged-polaron vertical single-points write to the Exp. 9 output archive under the open repository). Local dopant–cage metrics supporting the mechanistic synthesis are in Table S5.

COMPUTATIONAL METHODS (FULL PROTOCOL)

All total-energy and electronic-structure calculations use CP2K [2] with PBE [3] and Grimme DFT-D3 (Becke–Johnson damping) [4]. Unless stated otherwise, biaxial strain scales the in-plane lattice vectors of the pristine equilibrium cell at fixed fractional coordinates (rigid-strain protocol), providing an upper bound on strain–doping coupling until ionic-relaxation benchmarks in Table S3 converge. Plane-wave cutoff is 400 Ry for $n \leq 4$ and 350 Ry for $n \geq 6$ in archived Exp10 inputs (Table S2); $\mathcal{S}(n)$ comparisons at fixed n use identical cutoffs.

Tetramer models ($4 \times \text{C}_{60}$, 240 sites) use B/N/P substitution at nominal 5% (12–13 dopants from archived coordinates, seed 42). Periodic scaling cells contain one dopant per C₆₀ ($n = 1$ –8). Dopant sites are not re-optimized thermodynamically; fixed configurations probe mechanism rather than ensemble averages. A second random seed 137 tetramer grid (Table S4) bounds placement sensitivity.

Grids: 24 tetramer strain points ($\epsilon \in \{-5, -2.5, 0, 2.5, 5\}\%$); 12 tetramer electronic points; 40 converged periodic neutral-energy points (+1 cutoff400 control pending) at $\epsilon \in \{0, +3\}\%$; charged-state

optimizations for Marcus schematics (Fig. S5); IPR/ J factorial (Fig. S6). Linear $\alpha = dE/d\epsilon$ and substitution energies E_{sub} are defined in Table S1. **Task inventory (main-text support)**. Mechanical strain: 24 tetramer points ($\epsilon \in \{-5, -2.5, 0, 2.5, 5\}\%$, Table S1). Electronics: 12 tetramer PDOS/gap points ($\epsilon \in \{-5, 0, +5\}\%$, Fig. S4). Size scaling: 40 converged periodic neutral energies ($n = 1$ –8, $\epsilon \in \{0, +3\}\%$; production Exp10 grid; one matched 400 Ry $n=6$ N control pending, Table S2). Synergy $\mathcal{S}$ values are extracted from converged periodic total energies by in-house post-processing (Supplemental Table S2). Polaron factorial: four neutral single-points (Fig. S6); charged-state GEO-OPT (12/12 converged) and vertical SP (6/8 converged; 2/8 remaining) for Fig. S5. **Marcus vertical step**. Eight tetramer ENERGY tasks evaluate each adiabatic $q = \pm 1$ endpoint at the *neutral*-geometry atomic coordinates (400 Ry, PBE+D3, EPS_SCF 10^{-7} Eh), with UKS charge and multiplicity matched to the relaxed charged state (λ^+ from $q = +1$, λ^- from $q = -1$), following Ref. 1; $\lambda^\pm = E_{\text{vert}}^\pm - E_{\text{opt}}^\pm$ (Supplemental Fig. S5; verified values SI-only at revision). SCF: 10^{-6} Eh OT; basis DZVP-MOLOPT-SR-GTH. Pending ionic-relaxation, alternate-seed, and matched-cutoff controls are listed in Tables S2–S4.

Non-additive coupling metric. We quantify strain–doping cross terms with the synergy order parameter $\mathcal{S}$:

$$\mathcal{S}(\epsilon, \delta) = \Delta E(\epsilon, \delta) - [\Delta E(\epsilon, 0) + \Delta E(0, \delta)], \quad (\text{S1})$$

where ϵ is biaxial strain, $\delta \in \{B, N, P\}$, and $\Delta E(\epsilon, \delta)$ is the energy per atom relative to the pristine, zero-strain reference at the same supercell size $n \times \text{C}_{60}$. The bracketed term is the *additive* prediction from separate strain-only and doping-only scans; deviations from zero measure non-additive coupling. Additive superposition gives $\mathcal{S} = 0$; $\mathcal{S} < 0$ indicates cooperative stabilization beyond the two-scan sum, whereas $\mathcal{S} > 0$ indicates cooperative destabilization.

To first order in strain, $\mathcal{S}(\epsilon, \delta) \approx (\alpha_\delta - \alpha_0)\epsilon$ when $\alpha_\delta = dE/d\epsilon$ is evaluated for the doped cell and α_0 for the pristine reference at the same n (rigid-strain protocol); this links the mechanical coefficient α to the cross term at small ϵ .

Periodic-cell $\mathcal{S}$ values are extracted from converged CP2K total energies by in-house post-processing (Supplemental Material Table S2); the fifteen-point +3% strain audit is provided with the open data release. We report $\mathcal{S}$

TABLE S1. Total energies, linear strain sensitivities, and substitution energies for tetramer models (PBE+D3, rigid-strain protocol). $E_{\text{sub}} = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$ at $\epsilon=0$ with no chemical-potential reference (not a standard formation enthalpy). n_{dopants} is counted from archived input coordinates (nominal 5% target; B: 13, N: 12, P: 12).

System	n_{dop}	$E(0\%)$ (Ha)	$E(+5\%)$ (Ha)	α (meV/%)	E_{sub} (eV/dopant)
Pristine	—	-1363.5512	-1363.5509	-1.0	—
B-doped	13	-1328.4900	-1328.4090	+56.9	+73.4
N-doped	12	-1414.3812	-1414.4053	-303.2	-115.2
P-doped	12	-1370.4965	-1370.4888	-19.1	-3.8

for total energy only; an analogous gap cross term $\mathcal{S}_{\text{gap}}$ is not computed here—main-text panel (a) shows that gap slopes need not track $|\mathcal{S}|$.

Localization and coupling (Supplemental Material Fig. S6). Following Capobianco *et al.* [1], we report the inverse participation ratio (IPR) of the charged-state frontier orbital and the nearest-neighbor electronic coupling J from tetramer dimer fragments. IPR is $N \sum_i |\psi_i|^4 / (\sum_i |\psi_i|^2)^2$ on a Mulliken-projected orbital grid; J is estimated from the ground–first-excited splitting of the anionic dimer at fixed interfullerene geometry (two-state model). These observables are computed at the same PBE+D3 level as the energy scans and are *not* directly comparable in absolute value to the Koopmans-compliant $J \simeq 75\text{--}81$ meV reported for pristine qHP net-

works [1], but they provide an internally consistent trend under coupled (ϵ, δ) .

Polaron binding (Supplemental Material). Marcus reorganization energies follow the vertical-adiabatic decomposition [1]: $\lambda^\pm = E_{\text{vert}}^\pm - E_{\text{opt}}^\pm$, with separate internal ($\lambda_{\text{loc}}^\pm$) and environmental ($\lambda_{\text{env}}^\pm$) contributions when geometry subsets are relaxed. Charged-state relaxations and vertical single-points are documented in Supplemental Material Fig. S5. Twelve charged geometry optimizations supply adiabatic IP/EA; vertical single-points at +73.4 eV for neutral geometry are completing $\lambda^\pm$ (6/8 converged in 1st revision) (Supplemental Material Fig. S5).

TABLE S1

TABLE S2

TABLE S3

TABLE S4

TABLE S5

FIGURE S4

FIGURE S5

FIGURE S6

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- [1] A. Capobianco, A. Breglia, P. Gentile, G. Capano, A. Asinari, G. Drera, L. Sangaletti, M. Cossi, and S. Casassa, electron localization and mobility in monolayer fullerene networks, *Nano Letters* **24**, 7891 (2024).
- [2] J. VandeVondele, M. Krack, F. Mohamed, M. Parrinello, T. Chassaing, and J. Hutter, Quickstep: Fast and accurate density functional calculations using a mixed gaussian and plane waves approach, *Computer Physics Communications* **167**, 103 (2005).
- [3] J. P. Perdew, K. Burke, and M. Ernzerhof, Generalized gradient approximation made simple, *Physical Review Letters* **77**, 3865 (1996).
- [4] S. Grimme, S. Ehrlich, and L. Goerigk, Effect of the damping function in dispersion corrected density functional theory, *Journal of Computational Chemistry* **32**, 1456 (2011).

TABLE S2. Summary of converged DFT calculations and correspondence to main-text results. The ~ 90 tasks cited in the main text are the converged calculations that directly support quantitative claims; the full archive includes additional preliminary grids (rows marked “not used”).

Calculation set	Model	Converged tasks	Main-text usage	Notes
Early strain probe	$2 \times C_{60}$	5	not used	superseded by tetramer grids
Single-molecule doping	C_{60}	11	not used	superseded
Early electronics	$2 \times C_{60}$	24	not used	superseded by tetramer PDOS
Polaron IPR/ J factorial	tetramer	4	Fig. S6	pristine/ $B \times \epsilon=0, +3\%$
Tetramer strain energies	tetramer	24	Table S1, α	rigid strain
Optimal-design grid	tetramer	20	not used	archive only
Tetramer PDOS/gaps	tetramer	12	Fig. S4; main panels	
Geometry references	tetramer relax	6	charged-state inputs	reference geometries
Charged polarons	tetramer $q = \pm 1$	12/12 GEO_OPT	Fig. S5	6/8 vertical SP converged; verified $\lambda^\pm$ in Fig. S5 (pri
Size scaling	$n \times C_{60}$	40 (+1 pending)	main $\mathcal{S}(n)$	400 Ry ($n \leq 4$), 350 Ry ($n \geq 6$); cutoff400 control pe
$n=6$ cutoff control	$6 \times C_{60}$ N @+3%	pending	N sign benchmark	400 Ry matched control (pending)
Ionic relaxation (tetramer)	tetramer	pending	Table S3	fixed-cell GEO_OPT

TABLE S3. Ionic-relaxation benchmark for the tetramer $P@ + 3\%$ synergy corner (four fixed-cell GEO_OPT corners). Main-text $\mathcal{S}(n)$ in panel (d) of the composite figure uses *periodic* $n \times C_{60}$ rigid-strain energies (PBE+D3); this table tests whether a tetramer cross term survives atomic relaxation under the same biaxial cell. Rigid tetramer energies are from the archived tetramer strain-energy grid (PBE, no D3); relaxed energies use PBE+D3 fixed-cell GEO_OPT (300 Ry) as described in the Data Availability statement. $\mathcal{S}$ follows Eq. (S1) with $N=240$ atoms.

Quantity	Rigid (legacy tetramer)	Relaxed (GEO_OPT)
$E(\text{pristine}, \epsilon=0)$ (Ha)	−1363.5512	pending
$E(\text{pristine}, \epsilon=3\%)$ (Ha)	−1363.5522	pending
$E(P, \epsilon=0)$ (Ha)	−1370.4965	pending
$E(P, \epsilon=3\%)$ (Ha)	−1370.4890	pending
$\mathcal{S}$ (meV/atom)	+0.96	pending
Sign vs. periodic $n=4$ $\mathcal{S}_P$ (−23.7 meV/atom)	compare after convergence	

TABLE S4. Alternate random-seed dopant placement (tetramer, rigid strain, PBE+D3). Seed 42 is used in the main text; seed 137 repeats the full B/N/P strain grid ($\epsilon \in \{-5, -2.5, 0, +2.5, +3, +5\}\%$; 18 ENERGY tasks total) to test whether α and tetramer $\mathcal{S}$ at +3% are placement-sensitive. Periodic $n=4$ $\mathcal{S}$ values (seed 42) are listed for comparison only—different concentration and boundary conditions.

Dopant	Seed	α (meV/%)	$\mathcal{S}_{+3\%}^{\text{tet}}$ (meV/atom)	$\mathcal{S}_{+3\%}^{n=4}$ (meV/atom)	Status
B	42	+56.9	—	−6.2	verified (Table S1)
N	42	−303.2	—	+3.0	verified
P	42	−19.1	—	−23.7	verified
B	137	pending	pending	—	6 inp / pending
N	137	pending	pending	—	6 inp / pending
P	137	pending	pending	—	6 inp / pending

TABLE S5. Local dopant–cage structure from archived tetramer XYZ (mean nearest C distance $\bar{d}$ in Å; covalent-radius excess vs. C in pm). Strain-induced change $\Delta\bar{d} = \bar{d}(\epsilon = +3\%) - \bar{d}(\epsilon = 0)$ summarizes how tensile load couples to pre-existing distortion (see Discussion mechanistic synthesis). Values are computed from archived tetramer XYZ coordinates (seed 42) in the open data release; they do not include Mayer bond order or Bader charges.

Dopant	$\bar{d}(\epsilon=0)$ (Å)	$\bar{d}(\epsilon = +3\%)$ (Å)	$\Delta\bar{d}$ (Å)	Δr_{cov} (pm)
B	1.384	1.393	+0.0085	+7
N	1.393	1.402	+0.0087	−6
P	1.384	1.385	+0.00003	+30

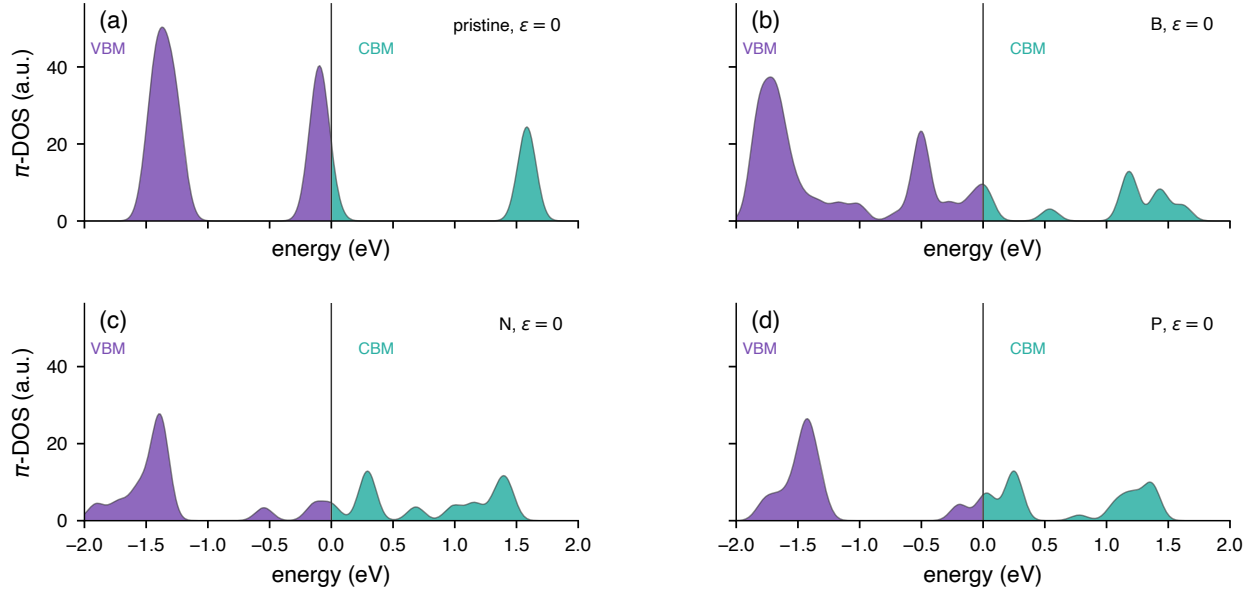


FIG. S1. π -projected density of states (π -DOS) for pristine and B/N/P-doped tetramers at $\epsilon=0$ (PBE+D3). Panels (a)–(d) share a common vertical scale for cross-dopant comparison. Valence band edges (purple) and conduction band edges (teal) follow the color scheme of Capobianco *et al.* [1]. Energy is relative to the Fermi level at 0 eV. Molecular-orbital isosurfaces are pending cube export.

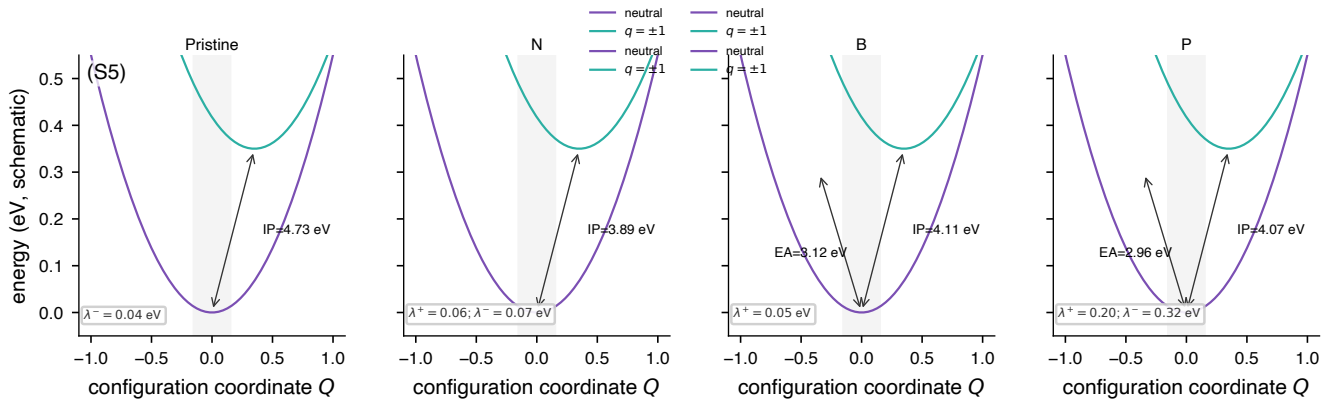


FIG. S2. Schematic configuration-coordinate diagrams for Marcus reorganization energies, following Ref. 1. Panels: pristine and N (hole, λ^+), B (electron, λ^- , IP+EA), and P (hole IP + EA; all four dopants from 12/12 charged GEO.OPT). Adiabatic ionization potential (IP) and electron affinity (EA) are from converged charged-state geometry optimizations (12/12 complete at revision); $\lambda^\pm = E_{\text{vert}}^\pm - E_{\text{opt}}^\pm$: pristine $\lambda^- = 0.042$, B $\lambda^+ = 0.051$, N $\lambda^+ = 0.058 / \lambda^- = 0.065$, P $\lambda^+ = 0.200 / \lambda^- = 0.315$ eV verified (6/8 vertical single-points converged); two tasks in progress at revision.

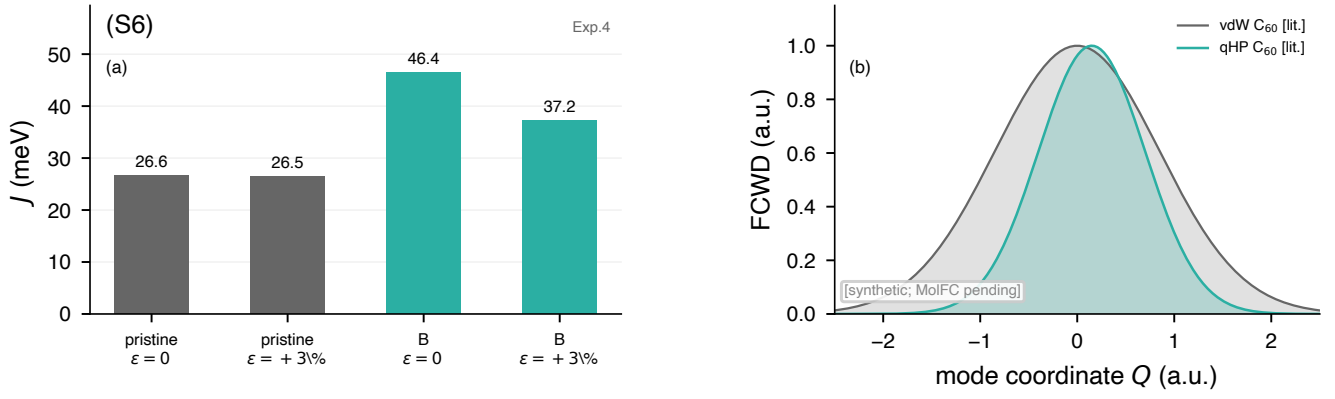


FIG. S3. Transport-adjacent observables (PBE+D3). **(a)** Nearest-neighbor electronic coupling J on the 2×2 factorial (pristine/B $\times \epsilon=0, +3\%$); single-point values without statistical error bars. **(b)** Illustrative Franck-Condon weighted density (FCWD) envelope—a *schematic placeholder* only; tetramer MolFC generating-function calculations are pending and this panel must not be read as a computed spectrum. The horizontal axis is a dimensionless mode coordinate Q .